

Separate Logs for Easier Troubleshooting

Question

The Operations team wants separate logs for easier troubleshooting.

Commands Used

```
sudo vi /etc/nginx/nginx.conf
sudo nginx -t
sudo systemctl reload nginx
curl -I http://localhost
curl http://localhost/nonexistent
sudo tail -n 10 /var/log/nginx/operations_access.log
sudo tail -n 10 /var/log/nginx/operations_error.log
sudo nginx -T 2>/dev/null | grep -E 'operations_(access|error)\.log'
```

Procedure

1. Opened the Nginx configuration file

```
sudo vi /etc/nginx/nginx.conf
```

2. Added separate access and error log configurations inside the http block

```
access_log /var/log/nginx/operations_access.log main;
error_log /var/log/nginx/operations_error.log notice;
```

The following separate log files were configured:

```
/var/log/nginx/operations_access.log
/var/log/nginx/operations_error.log
```

3. Tested the Nginx configuration

```
sudo nginx -t
```

The configuration test was successful.

4. Reloaded Nginx to apply the configuration

```
sudo systemctl reload nginx
```

5. Generated a successful HTTP request

```
curl -I http://localhost
```

This generated an entry in the separate access log.

6. Generated an actual Nginx error

```
curl http://localhost/nonexistent
```

This requested a nonexistent page and generated an error entry in the separate error log.

7. Verified the access log

```
sudo tail -n 10 /var/log/nginx/operations_access.log
```

8. Verified the error log

```
sudo tail -n 10 /var/log/nginx/operations_error.log
```

9. Verified that both log configurations were loaded by Nginx

```
sudo nginx -T 2>/dev/null | grep -E 'operations_(access|error)\.log'
```

Output

```
[ec2-user@ip-172-31-4-243 ~]$ echo "=== ACCESS LOG ==="
sudo tail -n 10 /var/log/nginx/operations_access.log
echo "=== ERROR LOG ==="
sudo tail -n 10 /var/log/nginx/operations_error.log
=== ACCESS LOG ===
::1 - - [10/Sep/2026:07:13:25 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.12.1" "-"
85.49.1.222 - - [10/Sep/2026:07:15:39 +0000] "\x16\x03\x01\x00{\x01\x00\x00w\x03\x03\x9eU\xce\xe72X\xba\x82\x9fm\x10 \xA9+\xA1\xbd[=\x9c\xA5\x0B\xD6i\xB3\xFB\x1Az\x1C\xA4k\xB5\xF0\x00\x00\x1A\xC0\xC0+\xC0+\xC0\x11\xC0\x07\xC0\x13\xC0\x09\xC0\x14\xC0" 400 157 "-" "-"
::1 - - [10/Sep/2026:07:15:47 +0000] "GET /nonexistent HTTP/1.1" 404 153 "-" "curl/8.12.1" "-"
::1 - - [10/Sep/2026:07:16:45 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.12.1" "-"
::1 - - [10/Sep/2026:07:16:45 +0000] "HEAD /nonexistent HTTP/1.1" 404 0 "-" "curl/8.12.1" "-"
=== ERROR LOG ===
2026/09/10 07:15:47 [error] 8396#8396: *181 open() "/usr/share/nginx/html/nonexistent" failed (2: No such file or directory), client: ::1, server: _, request: "GET /nonexistent HTTP/1.1", host: "localhost"
2026/09/10 07:16:45 [error] 8395#8395: *183 open() "/usr/share/nginx/html/nonexistent" failed (2: No such file or directory), client: ::1, server: _, request: "HEAD /nonexistent HTTP/1.1", host: "localhost"
[ec2-user@ip-172-31-4-243 ~]$
```

The access log contained HTTP request entries, while the error log contained the Nginx error generated by requesting the nonexistent page.

Result

Separate Nginx access and error logs were successfully configured and verified.

Conclusion

The Nginx web server was successfully configured with separate access and error log files. Real requests and an actual Nginx error were generated and recorded in the respective logs. This separate logging setup makes it easier for the Operations team to monitor requests and troubleshoot Nginx issues.